

11 Approximating Probabilities

- A probability can be interpreted as a theoretical long run relative frequency.
- The probability of event A can be approximated by simulating, according to the assumptions corresponding to the probability measure P , the random phenomenon a large number of times and computing the relative frequency of A .

$$P(A) \approx \frac{\text{number of repetitions on which } A \text{ occurs}}{\text{number of repetitions}}, \quad \text{for a large number of } P\text{-repetitions}$$

- In practice, many repetitions of a simulation are performed on a computer to approximate what happens in the “long run”. However, we often start by carrying out a few repetitions, (1) by hand to help make the process more concrete, or (2) by computer to check that our code is working correctly
- A probability can be approximated by a relative frequency from a large number of simulated repetitions, but there is some simulation margin of error, due to natural variability in the simulation.
- The margin of error when approximating a single probability based on a simulated relative frequency is roughly on the order $1/\sqrt{N}$, where N is the *number of independently simulated values used to calculate the relative frequency*.
 - Use a margin of error of $2/\sqrt{N}$ when approximating many probabilities based on a single simulation.
- Be careful when conditioning, especially when conditioning on values of continuous random variables.
- The larger N the more precise the estimate, but there is cost to simulating and storing more repetitions in terms of computational time and memory. More importantly, beware a false sense of precision: A precise estimate of a probability under the assumptions of the model is not necessarily a comparably precise estimate of the true probability.

Example 11.1. Roll a fair four-sided die twice and let X be the sum and Y the larger of the two rolls.

1. Explain how you would use simulation to approximate
 - a. $P(X = 6)$

- b. $P(X = 6, Y = 4)$
2. In 10000 repetitions of the simulation, $X = 6$ on 1915 repetitions. Approximate $P(X = 6)$, including a margin of error.
3. Explain how you would use simulation to approximate the distribution of X . What would the margin of error be?

Example 11.2. Recall the three person meeting problem in Example 10.2; let W be the time the first person waits for the last person.

1. Explain in full detail how you could conduct an appropriate simulation using spinners and use the results to approximate the probability $P(W < 15)$.
2. In 10000 simulated repetitions, $W < 15$ in 1821. Approximate and interpret $P(W < 15)$; include a margin of error.

Example 11.3. Consider simulating a randomly selected U.S. adult and determining whether or not the person has a college degree and whether or not they can fix a problem with a car's engine. Let C be the event that the selected adult has a college degree, E be the event that the selected adult can fix an engine, and P correspond to randomly selecting an American adult. Suppose that¹ $P(C) = 0.40$, $P(E) = 0.28$, and $P(C \cap E) = 0.07$.

1. In 10000 repetitions of an appropriate simulation

- C occurs on 3989 repetitions
- E occurs on 2837 repetitions
- Both C and E occur on 680 repetitions.

Use the simulation results to approximate $P(E|C)$.

2. What is the margin of error for your estimate in the previous part?

3. What is another method for performing the simulation and estimating $P(E|C)$ that has a margin of error of 0.01? What is the disadvantage of this method?

Example 11.4. Continuing Example 11.1, roll two fair four-sided dice and let X be the sum and Y the larger of the rolls.

1. For each of the following, describe how to approximate the probability with a margin of error of 0.01.

- a. $P(X = 6|Y = 4)$.

¹Values are based on this [Pew Research survey](#).

b. $P(Y = 4|X = 6)$.

2. Explain how to approximate the conditional distribution of X given $Y = 4$.

Example 11.5. Continuing Example 11.5. We want to approximate the conditional probability that $W < 15$ given that the first person arrives at 12:10. Explain in detail how you would conduct a simulation to approximate this probability. (Be careful!)